

TRISTAN DELA CRUZ

Irvine, California • 650-455-9078 • tristanmdelacruz@gmail.com • [Linkedin](#)

EDUCATION

University of California, Irvine - BS, Biomedical Engineering | GPA: 3.4

June 2028

WORK EXPERIENCE

TTM Technologies | PCB Manufacturing

Irvine, CA / JUNE 2026 - AUGUST 2026

Process Engineering/Manufacturing Intern

- Led three manufacturing initiatives to support process improvements from automation, DOEs, and quality systems
- Executed a Design of Experiments to determine quality parameters and features in a manufacturing process
- Implemented Ignition Automation to integrate sensors and a data history log to ensure accuracy and quality.

PROJECT EXPERIENCE

UCI HyperXite | Fail-Safe Pneumatic and Gas Spring Braking System

Irvine, CA / JUNE 2026 - Present

Braking Engineer

- Optimized a fail-safe braking system utilizing pneumatic actuators and gas springs to stop a 100kg hyper pod
- Simulated braking parameters using ANSYS Mechanical to ensure the quality and safety of the braking system
- Validated the braking system using in-house test rigs to test reliability and structural integrity of the braking system

UCI HyperXite | Spring and Damper Linkage Lateral Suspension System

Irvine, CA / JULY 2025 - JUNE 2026

Dynamics Engineer

- Redesigned a spring-damper suspension system that provides vehicle comfort, and optimized external input on pod
- Modeled vehicle dynamics using equations of motion in Simulink and Simscape for a high-speed HyperPod
- Assessed vehicle suspension system through MatLab and Simulink to ensure performance and response accuracy

UCI Legacy Robotics | Tank Steering Mars Rover Wheel Hub

Dynamics Engineer

Irvine, CA / JAN 2025 - SEPT 2025

- Spearheaded the design of an Independent steering wheel hub on a 100+ pound project rover for a rover competition
- Applied 3D Modeling using SolidWorks to fully design and manufacture a wheeled base to support a 100+ pound Rover
- Coordinated cross-communication effectively with 3+ interdepartmental teams to ensure design accuracy and quality

FUSION Engineering Project | Remote Controlled Precision Cargo Drone

Irvine, CA / DEC 2024 - MAY 2025

Mechanical Engineer

- Designed a Cargo Drone Frame and to be capable of transporting blocks and flying through competition hoops
- Executed 3D Printing process using Bambu Lab to manufacture and assemble drone modular components and claw fitting
- Delivered professional presentation skills through consistent design reviews to update our progress and gain feedback

FUSION Engineering Project | Mobile Gesture-Controlled Robotic Arm Rover

Irvine, CA / DEC 2023 - JUNE 2024

Dynamics Engineer

- Initiated the design of a Robotic Arm on a mini rover capable of transporting and stacking blocks on top of each other
- Utilized SolidWorks to design a 3-degree-of-freedom robotic arm, collaborating with statics and software subteams
- Demonstrated Finite Element Analysis (FEA) to identify issues, improvements, and adjustments during the design process

CERTIFICATIONS

Lean Six Sigma White Belt

TECHNICAL SKILLS

Design/Simulation: SOLIDWORKS, ANSYS, 3D Printing, 2D Drawings and Documents, Finite Element Analysis, MATLAB

Manufacturing: PCB Manufacturing, Design of Experiments (DOE), Metal Machinery, Woodworking Machinery

Software: Visual Studio Code, Ignition Automation, Python, HTML, Arduino, Exce